

Capnostat

1028494TL

Capno 5 External Power Supply, Test Fixture

File Approval

Responsible Engineer: Richard Daniels

Printed Name


Signature3.24.06

Date

Revision	C.O.	Date	Author	Description
51	N2919	08/04/05	Richard Daniels	Added schematic & assembly drawing.

1.0 PURPOSE

This document defines the external power supply (Respironics part number 1028494) for use with a Capnostat 5 Series Sensor.

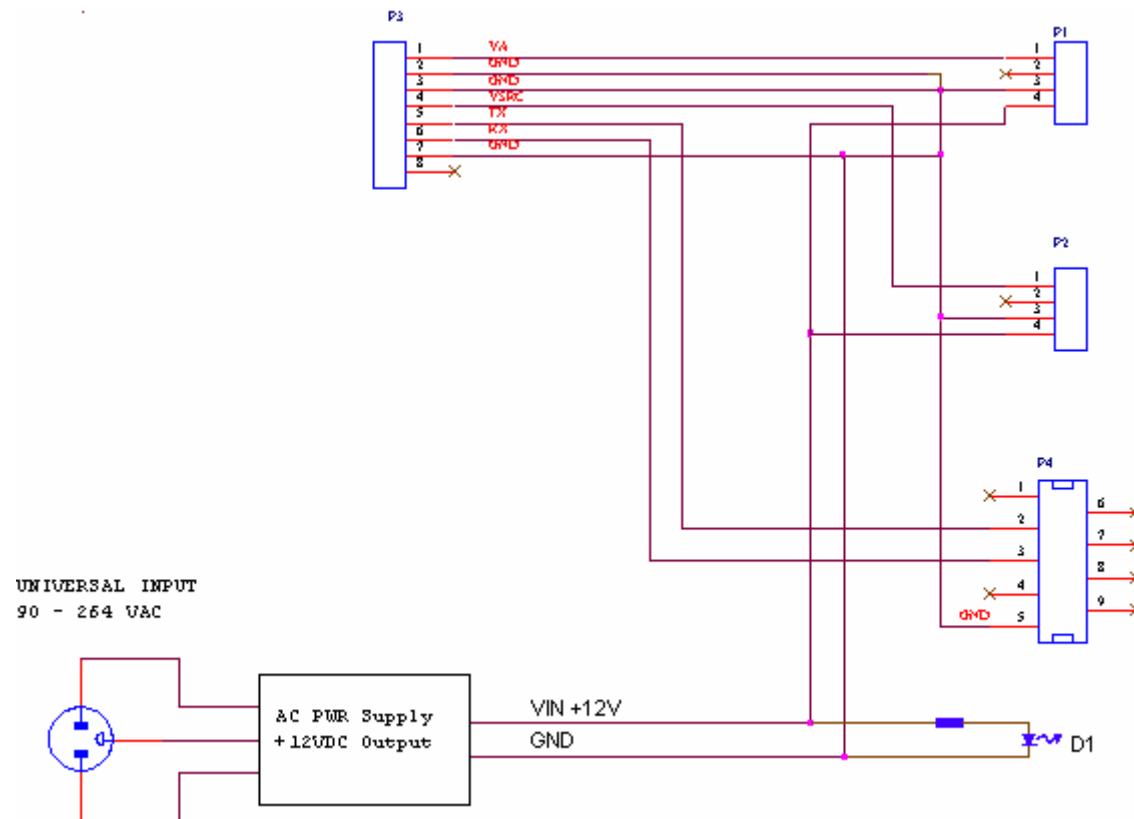
2.0 Theory

The fixture contains two identical +5VDC regulators. The external AC/DC adapter provides 12 VDC to the two PC boards. This voltage is regulated to +5VDC to provide power to the Capnostat 5 Sensor for its VA and VSRC power supply rails. Additionally, the RS-232 connection is converted to a standard 9 Pin D-Shell.

3.0 Assembly Bill Of Materials

Item	Quantity	Description	Part	MFG	MFG P/N	Respironics P/N
1	2	Buck SMPS Assembly	Supply PCB	N/A	N/A	2791-01 R51
2	1	LED	D1	Newark	87F937	
3	2	Connector, 4 position	P1, P2	AMP	640623-4	
4	1	UUT Receptacle	P3	LEMO	PKB.M0.8SL.LJ	
5	1	Connector, DB9	P4	N/A	N/A	211908
6	1	Screwlock (set)	HW3	AMP	207952-3	
7	1	Strain Relief	HW1	Eagle Plastic Devices	23MP02M40	
8	1	Enclosure	HW2	HAMMOND	1591LSBK	
9	1	AC PWR Supply	Supply, Desktop	Phihong	PSA-18U-120	
10	4	Rubber Feet	HW4	HAMMOND	1421T2	
11	1	Label	N/A	N/A	N/A	1023319 Rev 50

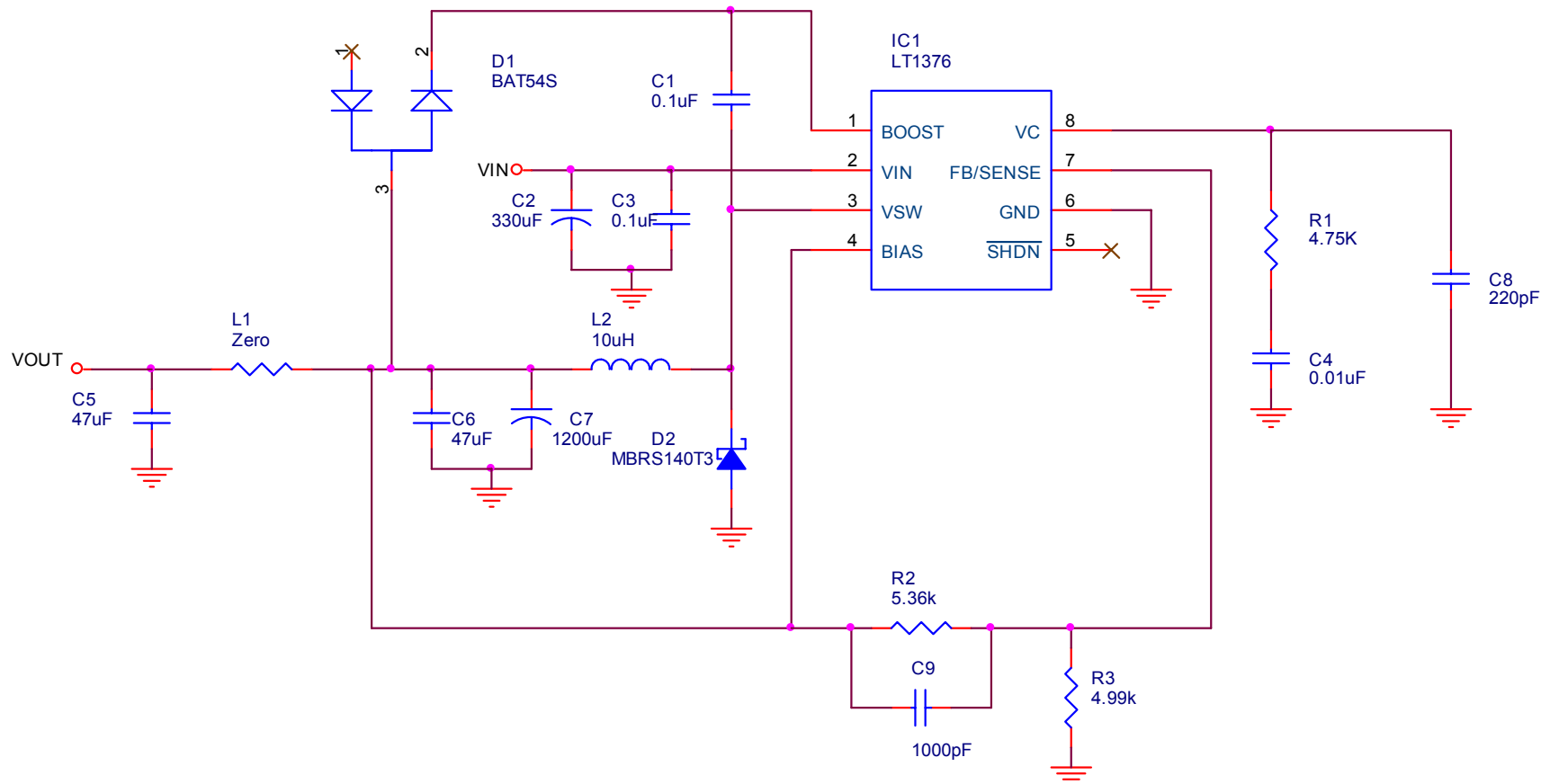
4.0 Wiring Diagram



5.0 Assembly Image 1028494



6.0 Reference Schematic – 2791-01 rev 51

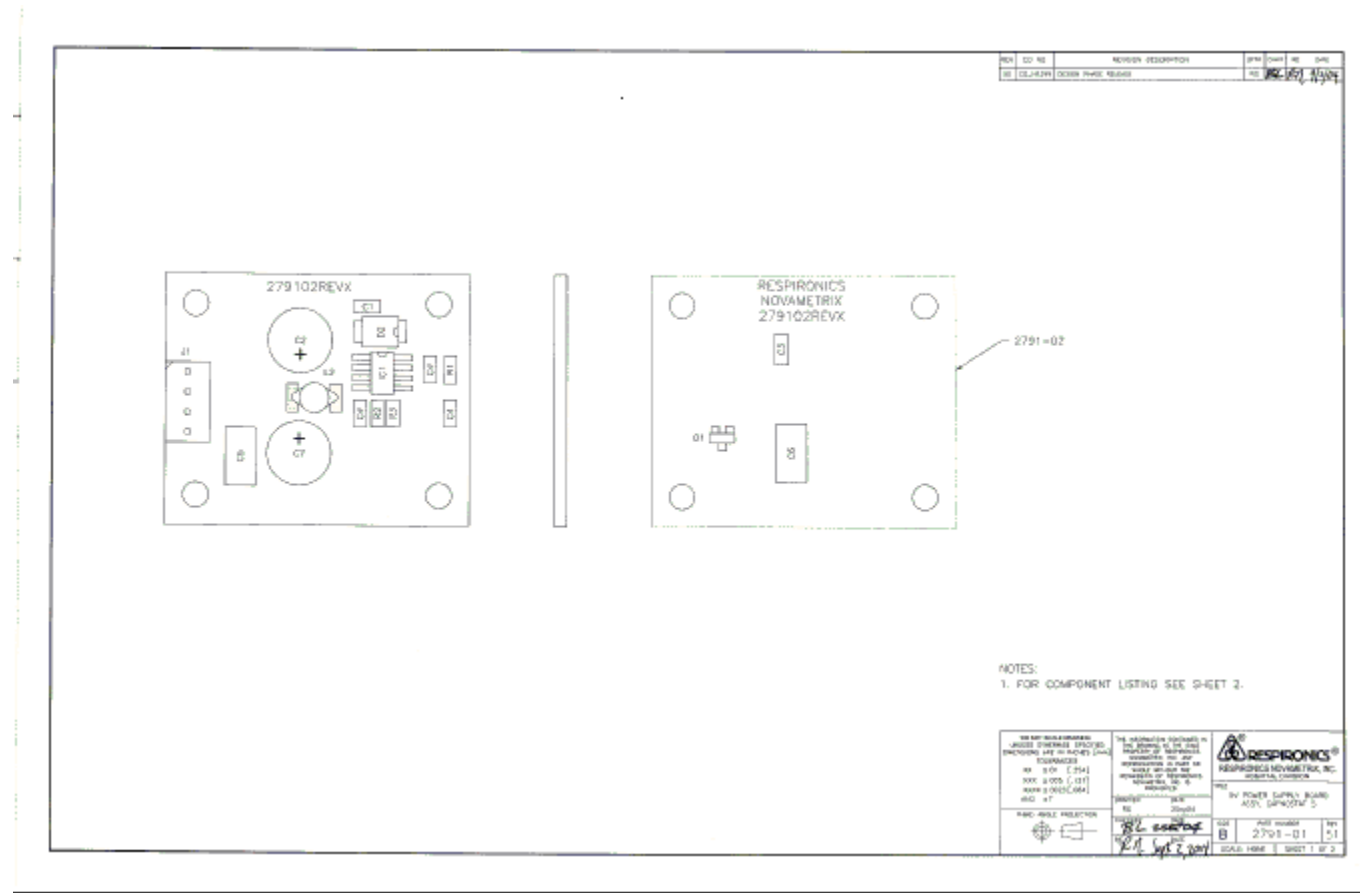


7.0 Reference Power Supply BOM 2791-01 Rev 51

5V POWER SUPPLY BOARD ASSY, CAPNOSTAT 5
2791-01rev51

Item	Quantity	Reference	Part	MFG	MFG P/N
1	1	C1	0.1uF	KEMET	C0603C104K3RACTU
2	1	C2	330uF, 25V	PANASONIC	EEU-FC1E331L
3	1	C7	1200uF, 6.3V	PANASONIC	EEU-FC0J122L
4	1	C3	0.1uF	PANASONIC	ECJ-2VB1E104K
5	2	C6,C5	47uF	PANASONIC	ECJ-5YB0J476M
6	1	D1	BAT54S	FAIRCHILD	BAT54S
7	1	D2	MBRS140T3	ON SEMI	MBRS140T3
8	1	IC1	LT1376	LINEAR TECH	LT1376CS8
9	1	J1	CON4	AMP	640452-4
10	1	L2	10uH	COILCRAFT	DO1608-103
11	1	R1	4.75k	PANASONIC	ERJ-3EKF4751V
12	1	R2	5.36k	PANASONIC	ERJ-3EKF5361V
13	1	R3	4.99k	PANASONIC	ERJ-3EKF4991V
14	1	N/A	Fab	N/A	2791p0
15	1	C4	0.01uF	KEMET	C0603C103K3RACTU
16	1	C8	220pf	PANASONIC	ECU-S1H221JCA
17	1	C9	1000pf	KEMET	C0603C102K3RACTU
18	1	L1	24 GA wire		

8.0 Reference Assembly 2791-01 Rev 51

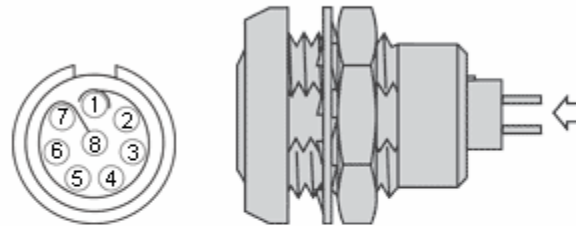


9.0 Reference Lemo Redel Receptacle

PERFORMANCE DETAILS

Insert Configuration 1P.M08: 8 Low Voltage

Insulator L: PEEK



Specifications

Low Voltage

Contact Type:	Solder
Max. Matings:	1000
Contact Dia.:	0.7 mm (0.0275")
Bucket Dia.:	0.6 mm (0.024")
Max. Solid Conductor:	0.14 mm ² (AWG 26)
Max. Stranded Conductor:	0.14 mm ² (AWG 26)
R (max):	6.5 mOhm
Vtest:	1050 V (AC), 1500 V (DC)
Rated Current:	5 A (AC/DC)

10.0 Calibration and Preventative Maintenance Schedule

Schedule

Calibration is not required because this fixture is not used to qualify product for use.

The Preventive Maintenance shall be performed every 12 months as defined in QAF81005.

11.0 Handling, Preservation, and Storage

There are no special handling and storage requirements for this tool.